Juan Vásquez

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Professional Summary

Backend software developer blending deep technical expertise in machine learning, cloud infrastructure, and system automation with extensive client-facing experience. Proficient in designing robust data pipelines, developing APIs, and managing Linux-based environments. Highly skilled at bridging the gap between complex backend systems, advanced analytics, and customer success.

Core Competencies & Top Skills

- **Languages & Frameworks:** Python (*PyTorch, Flask, FastAPI*), Rust (*Dioxus, Actix-Web*), SQL.
- **Cloud & Infrastructure:** Amazon Web Services (*AWS EC2, Lightsail, Lambda*), S3 Bucket Management, IAM Policies, Docker, Linux (*Debian, NixOS*).
- **Data & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Signal Processing (*CWT*), Extract, Transform, and Load Processes (*ETL*), Data Pipelines, Zoho Analytics.
- **Engineering, Integration & Delivery:** Backend Development, API Development, API Integration, Zapier, Workflow Automation, HelpScout, Customer Service.

Open Source Contributions

- Contributed to the Dioxus project, an open-source cross-platform framework for full-stack web, mobile, and desktop applications written in Rust. My contributions include addressing bug fixes and enhancing the parsing of Rust code into HTML.

Professional Experience

Spiro.AI (https://spiro.ai/) Caracas, Venezuela (Remote)
Customer Success Manager February 2024 - Present

- Developed custom automated workflows to handle the customer's data and internal processes.
- Created customized dashboards and reports tailored to support clients' Business Intelligence departments, leveraging platforms like Zoho Analytics.
- Developed advanced SQL queries to extract meaningful insights from complex datasets.
- Analyzed and mapped client data fields to ensure accurate integration into the Spiro system, including data sorting and restructuring.
- Managed and resolved client support tickets, ensuring timely and effective solutions to enhance customer satisfaction.
- Acted as a technical liaison between clients and the development team to report bugs, suggest improvements, and implement workarounds.
- Conducted weekly and monthly video meetings with clients to address issues, provide guidance, and answer questions regarding the Spiro app.

HI4AI Human Intelligence for Artificial Intelligence (https://hi4.ai/) Caracas, Venezuela (Remote)
Software Engineer March 2021 - February 2024

- Led multiple projects and proof of concepts (POCs) focused on leveraging technology to address client challenges through system integration and web application development.
- Deployed scalable solutions primarily on AWS (EC2, Lightsail, Lambda) and occasionally on Azure Virtual Machines, securing architecture through robust IAM policies and stringent S3 bucket owner enforcement.
- Developed a custom computer vision model to analyze aerial images, identifying items of interest on rooftops to assist clients in generating sales prospects.
- Built several tools to convert data from JSON, XML, or plain text formats into various file structures or directly into databases.
- Designed and implemented a web app for reviewing and approving installation reports, integrating data from the client's XOi account and facilitating the evaluation process by sending results to the client's Airtable account.
- Developed a custom keypoint detection model to evaluate hand-drawn sketches, ensuring they met specified criteria for validity.
- Created an automated tool to monitor Jira tasks, providing alerts based on specific conditions, along with a web app for managing tool activation and filter settings.
- Created a web app to analyze real estate market data, summarizing recent sales information for client decision-making.

- Utilized Python as the primary programming language and provided support in debugging JavaScript code.
- Gained experience in using Docker containers for application development.

VENPERCA Proyectos y Servicios Caracas, Venezuela
Drilling Specialist and Director July 2019 - September 2020

- Calculated head losses from wells to the main water line, selecting and installing appropriate well pumps to optimize water flow and efficiency.
- Designed and installed electrical control panels for well pumps and hydro-pneumatic systems, ensuring reliable operation and maintenance of water supply systems.
- Operated XY-3 well drilling machines to successfully drill drinking water wells, ensuring compliance with safety and quality standards.

Albatros Ingeniería Caracas, Venezuela
Projects Engineer and Director September 2008 - July 2019

- Designed and implemented a comprehensive maintenance plan for IT equipment, including the design and installation of internal network infrastructure, user configuration, and access permissions.
- Deployed virtual and physical servers to support various services, including Active Directory, file server, printer server, CRM, and ERP systems.
- Conducted computational simulations using Computational Fluid Dynamics (CFD) software, analyzing fluid flow in pipes, process equipment (both static and rotational), fluid separation and mixing, particle transport, external and free surface flows, open channel flow, and aerodynamic resistance of vehicles and wing profiles.
- Instructed basic and advanced courses on ANSYS CFX and ANSYS FLUENT software, enhancing team capabilities in fluid dynamics simulations.
- Supervised on-site assembly and erection of steel structures for housing complexes, ensuring compliance with engineering specifications and safety standards.
- Delivered tailored in-company sales presentations, addressing specific client needs and requirements.
- Maintained accurate accounting records, overseeing general bookkeeping, accounts payable and receivable, invoicing, and supplier payments.
- Prepared financial reports and coordinated with a certified public accountant for audits and tax declarations, ensuring compliance with government regulations.
- Managed general maintenance of office infrastructure, including furniture and equipment, coordinating the procurement of cleaning supplies and snacks for staff and visitors.

Empresas Y&V (https://www.yvsite.com/) Caracas, Venezuela
Projects Engineer April 2006 - September 2008

- Prepared data sheets, technical specifications, and calculation reports for pressure vessels in compliance with ASME Code, Section 8, Division 1, and atmospheric storage tanks according to API 650 standards.
- Conducted water hammer studies for piping systems to assess and mitigate pressure surge risks.
- Modeled static equipment using AutoPlant and Plant Design System (PDS), ensuring accurate representation of project specifications.
- Coordinated PDS activities, including initial project configuration, loading of piping components per project piping plans, and configuring orthographic and isometric drawings.
- Programmed EDEN modules for special components not available in PDS.
- Extracted reports for bills of materials using Material Data Publisher and Microsoft Access.
- Managed SmartPlant Instrumentation tasks, including importing instrument lists from Microsoft Excel and creating AutoCAD templates for the automated generation of Hook-Up plans.

Education

Universidad Simón Bolívar Caracas, Venezuela
Bachelor of Science (BS), Mechanical Engineering September 2000 – May 2006

Languages

Spanish (native) | English (advanced) | Portuguese (advanced) | Italian (beginner) | German (beginner)